



Accelerating configuration of Reconfigurable Intelligent Surfaces through a hardware-enhanced deep learning approach

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ABSTRACT

The dawn of Reconfigurable Intelligent Surfaces (RIS) promises to revolutionize wireless communication by enabling dynamic control of the propagation of electromagnetic waves. However, the practical implementation of RIS demands sophisticated configuration strategies to unlock their full potential. This paper presents a hardware implementation of a Deep Learning (DL) approach to the configuration of RIS, addressing both the complexity and efficiency of the configuration process. A deep learning-based algorithm, designed to optimize the phase shifts of RIS elements and shown to enhance signal quality and system performance, is implemented on two dedicated hardware platforms based on Field-Programmable Gate Arrays (FPGA), which leads to real-time processing and adaptability. This approach leverages the inherent parallelism of FPGAs to accelerate the computationally intensive tasks associated with deep learning inference. As a matter of fact, it is possible to achieve more than 18.000 configurations per second, thus ensuring rapid and efficient RIS configuration with a novel approach within this field.

1. Introduction

The rapid advancement of wireless communication technology necessitates innovative solutions to meet the ever-increasing demands for higher data rates, improved coverage, and energy efficiency. On June 2018, after years of research and development, the initial commercial roll-out of the 5th generation (5G) mobile communication standard was successfully completed. The new generation is already delivering speeds 10 times faster than its predecessor, with up to 20-Gbps peak data rates [1]. According to the Global System for Mobile Communications Association (GSMA), a non-profit organization representing the interest of mobile networks worldwide, 5G is on track to account for 15% of the global mobile industry by 2025 [2].

However, it has become clear during the standardization process of 5G wireless networks that no single enabling technology can meet all the application requirements of 5G [3]. Consequently, researchers have already begun exploring technologies beyond 5G

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